

1 Prognostic Nutritional Index and All-Cause Mortality in
2 Gastrointestinal Cancer: A Dual-Cohort NHANES
3 Analysis With Time-Dependent, Competing Risk, and
4 Joint Inflammatory Assessment

5 Yiqi Cai^b, Yongyu Bai^b, Qigang Xu^a, Zhuha Zhou^b

*^aDepartment of Hepatobiliary and Pancreatic Surgery, The First Affiliated Hospital of
Wenzhou Medical University, Wenzhou, China*

*^bDepartment of Gastroenterology Surgery, The First Affiliated Hospital of Wenzhou Medical
University, Wenzhou, China*

6 **Abstract**

7 **Background & Aims:** Nutritional status is a modifiable prognostic factor
8 in cancer patients, but population-level validation of composite nutritional in-
9 dices in gastrointestinal (GI) cancer remains limited. We aimed to validate the
10 Prognostic Nutritional Index (PNI), Controlling Nutritional Status (CONUT),
11 and Geriatric Nutritional Risk Index (GNRI) as prognostic factors for all-cause
12 mortality in GI cancer patients using nationally representative data.

13 **Methods:** We identified 313 GI cancer patients from NHANES III (1988–
14 1994, n=92) and Continuous NHANES (2005–2016, n=221) with mortality
15 follow-up through December 2019 (up to 31 years). PNI, CONUT, and GNRI
16 were derived from baseline serum albumin, lymphocyte count, total cholesterol,
17 and body mass index. Cox proportional hazards, time-dependent Cox, landmark
18 analysis, competing risks, restricted cubic splines, restricted mean survival time
19 (RMST), and population attributable fraction (PAF) analyses were performed.
20 E-value sensitivity analysis quantified robustness to unmeasured confounding.

21 **Results:** Higher PNI was associated with reduced all-cause mortality (ad-
22 justed HR 0.78, 95% CI 0.63–0.96, p=0.020). GNRI showed the strongest associ-
23 ation (HR 0.67, 95% CI 0.57–0.80, p<0.001). The PNI effect was strongly time-

*Corresponding author.

**These authors contributed equally to this work and share first authorship.
Email address: zhouzhuha@wmu.edu.cn (Zhuha Zhou)

24 dependent, maximal in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$)
 25 and attenuating thereafter (interaction $p < 0.001$). Optimal PNI threshold was
 26 48.5 (cross-validated mean 48.3); patients below this had median survival 6.6
 27 vs 12.8 years ($p < 0.0001$) with RMST difference 1.97 years (95% CI 1.19–2.76,
 28 $p < 0.001$). In competing risks, PNI primarily protected against non-cancer death
 29 (HR 0.63, $p = 0.001$) rather than GI cancer-specific death (HR 1.06, $p = 0.70$).
 30 PNI decomposition showed albumin alone outperformed the composite index
 31 ($\Delta C\text{-stat} +0.032$ vs $+0.022$), while lymphocyte count contributed no indepen-
 32 dent prognostic information (HR 1.06, $p = 0.54$). Dietary quality (HEI-2015)
 33 did not predict survival (HR 0.97, $p = 0.85$). Inflammatory markers (NLR,
 34 SII) showed comparable prognostic performance to PNI. Results were consis-
 35 tent across NHANES generations (cohort interaction $p = 0.48$) and robust to
 36 adjustment for education, income, smoking, and comorbidity. Fine-Gray sub-
 37 distribution hazard models confirmed the competing risks patterns. CONUT
 38 and GNRI survived Bonferroni correction for multiple testing ($\alpha = 0.0083$),
 39 while PNI (adjusted $p = 0.020$) did not.

40 **Conclusions:** PNI, CONUT, and GNRI are robust, independent prognos-
 41 tic factors for all-cause mortality in GI cancer patients, validated across two
 42 nationally representative US cohorts spanning 30 years. The protective effect
 43 is time-dependent, operates primarily through non-cancer mortality pathways,
 44 and is driven by serum albumin. These simple, inexpensive indices may guide
 45 early nutritional intervention in GI cancer patients.

46 *Keywords:* Prognostic Nutritional Index, Gastrointestinal Cancer, NHANES,
 47 Survival Analysis, Competing Risks, Time-Dependent Effect

48 1. Introduction

49 Cancer cachexia and malnutrition represent critical challenges in gastroin-
 50 testinal oncology. GI cancer patients are uniquely vulnerable due to tumor-

51 induced catabolism, malabsorption, and mechanical obstruction, with malnutri-
52 tion prevalence ranging from 30% to 80% depending on tumor site and stage [1,
53 2, 3]. Malnourished patients consistently show poorer treatment tolerance, in-
54 creased postoperative complications, and reduced survival [4, 5].

55 Several composite nutritional indices have been developed to integrate multi-
56 ple biomarkers into a single prognostic score. The Prognostic Nutritional Index
57 (PNI), originally proposed by Onodera et al. in 1984, combines serum albu-
58 min and peripheral lymphocyte count [6]. The Controlling Nutritional Sta-
59 tus (CONUT) score incorporates albumin, lymphocyte count, and total cholest-
60 erol [7]. The Geriatric Nutritional Risk Index (GNRI) adjusts albumin for body
61 mass index [8]. Each index has been validated primarily in single-center surgi-
62 cal series, predominantly in East Asian populations with evidence concentrated
63 in specific GI cancer subtypes such as gastric, colorectal, and hepatocellular
64 carcinoma [9, 10, 11].

65 Despite their clinical utility, important gaps remain. First, population-level
66 validation across different geographic contexts and survey eras is lacking. Most
67 prior studies originate from single-institution cohorts with sample sizes rarely
68 exceeding 1,000 patients [12, 13, 14]. Second, the temporal dynamics of nutri-
69 tional risk—whether the prognostic effect is constant or concentrated in specific
70 post-diagnosis windows—have not been systematically examined using time-
71 dependent analyses. Third, the conceptual distinction between biochemical
72 nutritional status (serum proteins, lymphocyte count) and dietary nutritional
73 quality (nutrient intake patterns) remains poorly characterized in cancer prog-
74 nosis.

75 To address these gaps, we undertook a comprehensive analysis of nutritional
76 indices in GI cancer patients using the National Health and Nutrition Exami-
77 nation Survey (NHANES), a nationally representative population-based survey

with mortality linkage. We leveraged two independent NHANES generations—
NHANES III (1988–1994) and Continuous NHANES (2005–2016)—providing
up to 31 years of mortality follow-up and enabling cross-validation across different
survey eras. Our objectives were to: (1) validate PNI, CONUT, and GNRI
as prognostic factors for all-cause mortality; (2) characterize time-dependent effects
through landmark, time-dependent Cox, and RMST analyses; (3) compare
prognostic performance across competing causes of death; and (4) compare biochemical
nutritional indices against a dietary quality measure (HEI-2015) and inflammatory
markers (NLR, SII).

2. Materials and Methods

2.1. Study Population

We used data from two independent NHANES cycles: NHANES III (1988–1994) and Continuous NHANES (2005–2016). NHANES is a nationally representative cross-sectional survey of the civilian non-institutionalized US population conducted by the National Center for Health Statistics (NCHS). Both surveys collected demographic, dietary, laboratory, and examination data through standardized protocols. The analytic sample included participants aged ≥ 18 years with available nutrition biomarker data. A total of 44,066 participants met inclusion criteria, of whom 313 had a GI cancer diagnosis.

The NHANES protocol was approved by the NCHS Ethics Review Board (Protocol #2005-06 for 2005–2016 cycles; Protocol #98-12 and Institutional Review Board approval for NHANES III). Written informed consent was obtained from all participants. As a secondary analysis of de-identified public-use data, this study was exempt from additional institutional review board approval.

102 2.2. GI Cancer Ascertainment

103 GI cancer was identified from self-reported cancer history. In Continuous
104 NHANES (2005–2016), cancer diagnoses were identified through the Medical
105 Conditions Questionnaire (MCQ) using cancer site codes. In NHANES III,
106 cancer site was identified from variable HAC3OS, with GI sites defined as codes
107 for colon, stomach, esophagus, pancreas, liver, and gallbladder. GI cancer was
108 defined as cancer of the esophagus, stomach, colon, rectum, pancreas, liver, or
109 gallbladder.

110 2.3. Nutritional Indices

111 **PNI** was calculated according to Onodera’s formula: $PNI = 10 \times \text{serum}$
112 $\text{albumin (g/dL)} + 0.005 \times \text{lymphocyte count } (\mu\text{L}^{-1})$ [6].

113 **CONUT** was derived as the sum of three component scores: albumin (≥ 3.5
114 $\text{g/dL} = 0$, $3.0\text{--}3.49 = 2$, $2.5\text{--}2.99 = 4$, $< 2.5 = 6$), lymphocyte count (≥ 1600
115 $\mu\text{L}^{-1} = 0$, $1200\text{--}1599 = 1$, $800\text{--}1199 = 2$, $< 800 = 3$), and total cholesterol (≥ 180
116 $\text{mg/dL} = 0$, $140\text{--}179 = 1$, $100\text{--}139 = 2$, $< 100 = 3$) [7].

117 **GNRI** was calculated as: $GNRI = 14.89 \times \text{albumin (g/dL)} + 41.7 \times (\text{BMI}$
118 $/ 22)$ [8].

119 Serum albumin was measured using the bromcresol purple method in Con-
120 tinuous NHANES and bromcresol green in NHANES III, with g/L values con-
121 verted to g/dL for consistency. Complete blood count provided lymphocyte
122 counts. Total cholesterol was measured enzymatically.

123 **HEI-2015** was calculated from 24-hour dietary recall data using the Food
124 Patterns Equivalents Database, measuring adherence to the 2015–2020 Dietary
125 Guidelines for Americans on a 0–100 scale.

126 2.4. Mortality Linkage

127 Mortality status was ascertained through linkage to the NCHS National
128 Death Index through December 31, 2019. Follow-up time was calculated from

the date of survey participation to death or censoring. The underlying cause of death was classified using the UCOD_LEADING variable, with malignant neoplasms (code 002) indicating cancer death.

2.5. Covariates

Demographic covariates included age (continuous, years), sex (male/female), and race/ethnicity (Non-Hispanic White, Non-Hispanic Black, Hispanic, Other), selected a priori based on known associations with both nutritional status and mortality. Extended models additionally adjusted for education (high school or above), income-to-poverty ratio, and physical activity (any moderate-to-vigorous activity). Cohort (NHANES III vs. Continuous NHANES) was included as a fixed effect to account for differences in albumin assay methodology between survey eras.

2.6. Statistical Analysis

All analyses were conducted using R 4.6.0 with the `survey`, `survival`, `EValue`, `survRM2`, and `broom` packages.

Survival analysis: Cox proportional hazards models estimated hazard ratios (HR) per 1-standard deviation (SD) increase in each nutritional index. Both crude and adjusted models (age, sex, race/ethnicity) were fitted. The proportional hazards assumption was tested using Schoenfeld residuals. For PNI, which violated this assumption ($p=0.017$), time-dependent Cox models were fitted with time splitting at 2 and 5 years.

Time-dependent effects: Landmark analyses at 1, 3, and 5 years examined conditional survival. Restricted mean survival time (RMST) at 10 years was compared between high and low PNI groups using the threshold identified by maximally selected log-rank statistics [15, 16]. Restricted cubic splines (3 knots) examined nonlinear dose-response relationships.

155 **Competing risks:** Cause-specific Cox models were fitted for GI cancer
156 death vs. non-cancer death [17, 18]. Subdistribution hazard models (Fine-Gray)
157 were additionally fitted to address the competing risk structure from an absolute
158 risk perspective [19].

159 **Component analysis:** PNI was decomposed into albumin alone and lym-
160 phocyte alone, comparing their prognostic performance to the composite index
161 using AIC, change in C-statistic with 2000 bootstrap replications, and likelihood
162 ratio tests.

163 **Cohort-stratified and interaction analyses:** Cohort-stratified models
164 (NHANES III vs. Continuous NHANES) examined effect consistency. Formal
165 cohort-by-index interaction terms tested heterogeneity. Cohort was also in-
166 cluded as a fixed effect to address albumin assay differences (bromocresol green
167 in NHANES III vs. bromocresol purple in Continuous NHANES).

168 **Extended covariate models:** Sensitivity analyses additionally adjusted
169 for education (high school or above), income-to-poverty ratio, smoking status
170 (current vs. non-current), and a simplified Charlson comorbidity score (myocar-
171 dial infarction, congestive heart failure, stroke, chronic obstructive pulmonary
172 disease, liver disease, and diabetes) to assess robustness to socioeconomic, be-
173 havioral, and comorbidity confounding. Multiple testing was addressed using
174 Bonferroni correction for the primary family of six tests (three indices $\times$ two
175 adjustment levels).

176 **Reverse causation sensitivity:** Analyses excluding deaths within the first
177 2 years of follow-up addressed potential reverse causation from undiagnosed
178 advanced disease at baseline.

179 **Joint exposure analysis:** PNI and NLR were dichotomized at medians
180 into four groups, with additive interaction quantified using Relative Excess Risk
181 due to Interaction (RERI) with bootstrap 95% CI.

182 **Population attributable fraction (PAF):** $PAF = p \times (HR - 1) / HR$
183 was estimated for $PNI < 48.5$ with bootstrap CI [20].

184 **E-value sensitivity analysis:** E-values quantified the minimum associa-
185 tion strength an unmeasured confounder would need to fully explain away the
186 observed results [21].

187 **Bootstrap C-statistic:** Incremental discrimination (ΔC -stat) over base-
188 line covariates was estimated with 500 bootstrap replications.

189 **Inflammatory markers:** NLR and SII were compared with PNI in the
190 2005–2016 subset with available neutrophil and platelet counts.

191 **3. Results**

192 *3.1. Baseline Characteristics*

193 Of 44,066 eligible participants aged ≥ 18 , 313 (0.7%) had a GI cancer di-
194 agnosis. Colorectal cancer was the most common (n=241, 77.0%), followed by
195 liver (n=23, 7.3%), stomach (n=19, 6.1%), esophageal (n=15, 4.8%), pancre-
196 atic (n=8, 2.6%), and gallbladder (n=6, 1.9%). The remaining cases were rectal
197 (n=8). GI cancer patients were substantially older than the general popula-
198 tion (67.5 vs 45.5 years) and had lower PNI (50.8 vs 53.2), indicating worse
199 nutritional status. During a median follow-up of 8.8 years (up to 31 years), 169
200 deaths (54.0%) occurred among GI cancer patients.

201 Table 1 summarizes baseline characteristics by GI tumor status, and Table 2
202 shows characteristics by PNI tertile among GI cancer patients. Patients in the
203 lowest PNI tertile had substantially higher mortality (70.5%) compared to the
204 middle (46.2%) and highest (45.2%) tertiles, with median survival of 6.5, 12.5,
205 and 13.9 years, respectively.

206 3.2. All-Cause Mortality

207 All three nutritional indices significantly predicted all-cause mortality af-
 208 ter adjustment for age, sex, and race/ethnicity (Table 3). GNRI showed the
 209 strongest association (HR 0.67 per 1-SD, 95% CI 0.57–0.80, $p < 0.001$), followed
 210 by CONUT (HR 0.71, 95% CI 0.61–0.83, $p < 0.001$) and PNI (HR 0.78, 95% CI
 211 0.63–0.96, $p = 0.020$).

Table 1: Cox proportional hazards—nutritional indices and all-cause mortality.

Index	Adjustment	HR (95% CI) per 1-SD	p-value
PNI	Crude	0.63 (0.51–0.78)	< 0.001
PNI	Adjusted*	0.78 (0.63–0.96)	0.019
CONUT	Crude	0.63 (0.55–0.73)	< 0.001
CONUT	Adjusted*	0.71 (0.61–0.83)	< 0.001
GNRI	Crude	0.67 (0.56–0.79)	< 0.001
GNRI	Adjusted*	0.67 (0.57–0.80)	< 0.001

*Adjusted for age, sex, race/ethnicity.

212 3.3. Time-Dependent Effects

213 The proportional hazards assumption was violated for PNI (Schoenfeld resid-
 214 uals $p = 0.017$) but not for GNRI ($p = 0.087$) or CONUT ($p = 0.43$). Time-dependent
 215 Cox models revealed marked attenuation of the PNI effect: the protective asso-
 216 ciation was strongest in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$)
 217 and non-significant thereafter (2–5 year: HR 1.32, $p = 0.49$; 5+ year: HR 0.81,
 218 $p = 0.45$; interaction $p < 0.001$). GNRI showed a similar temporal pattern (0–2
 219 year HR 0.53, $p = 0.034$), while CONUT demonstrated more stable effects across
 220 periods.

221 In landmark analysis, PNI showed attenuated and non-significant effects at
 222 1, 3, and 5 years, while CONUT and GNRI maintained significant associations
 223 at all landmark times (Table 4).

Table 2: Landmark analysis—adjusted HR per 1-SD.

Index	1-year Landmark	3-year Landmark	5-year Landmark
PNI	0.91 (0.73–1.13, p=0.38)	0.85 (0.66–1.10, p=0.21)	0.81 (0.60–1.09, p=0.17)
CONUT	0.75 (0.63–0.89, p<0.001)	0.71 (0.58–0.88, p=0.002)	0.74 (0.57–0.96, p=0.021)
GNRI	0.70 (0.59–0.84, p<0.001)	0.70 (0.57–0.86, p<0.001)	0.70 (0.56–0.89, p=0.003)

3.4. Optimal PNI Threshold and Dose-Response

Maximally selected log-rank statistics identified **PNI < 48.5** as the optimal prognostic threshold (p<0.0001). Patients below this threshold had markedly shorter median survival (6.6 vs 12.8 years). RMST analysis at 10 years confirmed a clinically meaningful difference of 1.97 years between groups (95% CI 1.19–2.76, p<0.001), with PAF indicating 13.4% of deaths attributable to low PNI (95% CI 6.2%–20.1%).

Restricted cubic spline analysis confirmed a nonlinear relationship between PNI and mortality risk (nonlinearity p<0.0001), with sharp risk increase below approximately 45 and a plateau above 50. In contrast, CONUT and GNRI showed linear dose-response relationships (nonlinearity p=0.11 and p=0.16, respectively).

3.5. Competing Risks Analysis

Cause-specific Cox models revealed divergent patterns. PNI primarily protected against non-cancer death (HR 0.63, 95% CI 0.48–0.83, p<0.001) rather than GI cancer-specific death (HR 1.06, 95% CI 0.78–1.45, p=0.70). CONUT and GNRI showed protective effects against both cancer-specific and non-cancer death (Table 5).

Fine-Gray subdistribution hazard models [19] confirmed these patterns. For GI cancer death, no index reached statistical significance (PNI sHR 1.16, 95% CI 0.77–1.75, p=0.49; CONUT sHR 0.82, 95% CI 0.59–1.14, p=0.24; GNRI sHR 0.79, 95% CI 0.58–1.07, p=0.12). For non-cancer death, GNRI remained

Table 3: Competing risks—cause-specific HR per 1-SD.

Index	GI Cancer Death	Non-Cancer Death
PNI	1.06 (0.78–1.45, p=0.70)	0.63 (0.48–0.83, p<0.001)
CONUT	0.73 (0.56–0.94, p=0.016)	0.70 (0.58–0.85, p<0.001)
GNRI	0.69 (0.51–0.94, p=0.017)	0.66 (0.53–0.82, p<0.001)

significant (sHR 0.80, 95% CI 0.64–0.99, p=0.039) and PNI was borderline (sHR 0.81, 95% CI 0.66–1.00, p=0.050).

3.6. PNI Decomposition

Albumin alone was significantly associated with mortality (HR 0.69 per 1-SD, 95% CI 0.58–0.82, p<0.001), while lymphocyte count alone was not (HR 1.06, 95% CI 0.88–1.28, p=0.54). These findings are consistent with prior systematic reviews demonstrating that low serum albumin is a robust predictor of cancer mortality across multiple tumor types [22, 23]. C-statistic improvement over baseline was +0.033 for albumin, +0.023 for PNI, and 0.000 for lymphocyte, confirming that **serum albumin is the primary driver** of PNI’s prognostic value.

3.7. PNI vs Inflammatory Markers

In the subset with neutrophil and platelet counts (n=258, 125 events), NLR (HR 1.43 per 1-SD, 95% CI 1.19–1.72, p<0.001) and SII (HR 1.30, 95% CI 1.10–1.54, p=0.002) significantly predicted mortality. These findings are consistent with prior meta-analyses demonstrating NLR as a robust prognostic marker across solid tumors [24]. C-statistic improvement was comparable across indices (+0.017 for PNI, +0.017 for logNLR). In mutually adjusted models including both PNI and logNLR, neither remained significant (PNI HR 0.86, p=0.29; logNLR HR 1.22, p=0.09), indicating shared predictive information.

266 3.8. Joint PNI $\times$ NLR Analysis

267 The “Low PNI + High NLR” group had significantly higher mortality (HR
268 1.67, 95% CI 1.06–2.63, $p=0.028$) compared to the “High PNI + Low NLR” ref-
269 erence group. RERI was -0.97 (95% CI -2.61 to -0.10), suggesting overlapping
270 rather than synergistic pathways.

271 3.9. CRP Interaction and CALLY Index

272 Among 250 patients with CRP measurements, CRP was non-significantly
273 associated with mortality (HR 1.18 per log-SD, $p=0.19$). No PNI $\times$ CRP in-
274 teraction was observed ($p=0.78$). The CALLY index (albumin $\times$ lymphocyte /
275 CRP) did not outperform PNI (CALLY HR 0.85, $p=0.20$; C-stat 0.659 vs 0.667
276 for PNI), despite prior studies reporting its prognostic value in colorectal cancer
277 populations [25].

278 3.10. Dietary Quality Comparison

279 Among 99 GI cancer patients with dietary data, HEI-2015 did not predict
280 mortality (HR 0.97, $p=0.85$). The correlation between HEI-2015 and PNI was
281 weak ($r=0.16$), confirming these measures capture distinct dimensions of nutri-
282 tional status.

283 3.11. Subgroup and Sensitivity Analyses

284 PNI showed directionally consistent protective effects across all examined
285 subgroups, with no significant effect modification by age (interaction $p=0.73$),
286 sex ($p=0.34$), or BMI ($p=0.68$). Bootstrap C-statistic (500 replications) showed
287 GNRI had the largest Δ C-stat ($+0.034$), followed by CONUT ($+0.023$) and
288 PNI ($+0.022$). Survey-weighted Cox (2005–2016 subset) attenuated PNI (HR
289 0.87, $p=0.43$) while CONUT remained significant (HR 0.76, $p=0.003$). E-values
290 ranged from 1.66 (PNI) to 1.97 (GNRI), indicating moderate robustness to
291 unmeasured confounding.

292 *3.12. Cohort Consistency and Extended Sensitivity Analyses*

293 Cohort-stratified analyses revealed consistent effects across NHANES gen-
294 erations. PNI showed comparable associations in NHANES III (HR 0.69, 95%
295 CI 0.51–0.93, $p=0.015$) and Continuous NHANES (HR 0.75, 95% CI 0.56–1.00,
296 $p=0.047$), with no significant cohort-by-PNI interaction ($p=0.48$). Similar con-
297 sistency was observed for CONUT (interaction $p=0.94$) and GNRI (interaction
298 $p=0.95$), supporting the appropriateness of pooling these cohorts. Inclusion of
299 cohort as a fixed effect did not materially alter the results (e.g., PNI HR 0.73,
300 95% CI 0.59–0.89).

301 Adjustment for additional socioeconomic covariates (education, income-to-
302 poverty ratio) modestly strengthened the PNI association (HR 0.74, 95% CI
303 0.55–1.00, $p=0.047$) and did not attenuate CONUT or GNRI. Further adjust-
304 ment for smoking status and Charlson comorbidity score in the Continuous
305 NHANES subset with available data ($N=133$, 61 events) did not attenuate the
306 associations: PNI HR 0.68 (95% CI 0.48–0.97, $p=0.035$), CONUT HR 0.63
307 (95% CI 0.50–0.80, $p<0.001$), and GNRI HR 0.72 (95% CI 0.55–0.93, $p=0.013$).
308 The stability of the effect estimates across progressive adjustment for socioe-
309 conomic, behavioral, and comorbidity confounders strengthens the evidence for
310 independent prognostic value, though residual confounding by unmeasured fac-
311 tors remains possible.

312 In sensitivity analyses excluding deaths within 2 years ($n=284$, 140 events),
313 PNI was no longer associated with mortality (HR 0.98, 95% CI 0.78–1.23,
314 $p=0.83$), confirming that its prognostic effect is concentrated in the early post-
315 assessment period. In contrast, CONUT and GNRI maintained significant as-
316 sociations (HR 0.77, $p=0.006$ and HR 0.70, $p<0.001$, respectively), suggesting
317 more stable long-term prognostic value.

3.13. Multiple Testing Consideration

In the primary analysis family of six tests (three indices $\times$ crude/adjusted models), the Bonferroni-corrected significance threshold was $\alpha = 0.0083$. PNI adjusted ($p=0.020$) did not survive this correction, while CONUT and GNRI adjusted (both $p<0.001$) and all crude models (all $p<0.001$) remained significant. These results are therefore reported as primary analyses for CONUT and GNRI, with PNI interpreted as a secondary finding requiring replication.

3.14. PNI Decomposition: Albumin vs. Composite Index

Formal comparison of PNI against its individual components confirmed that the prognostic value was driven entirely by serum albumin. Albumin alone achieved a lower AIC (1595.8 vs. 1607.8) and larger bootstrap C-statistic improvement ($\Delta C\text{-stat} +0.032$, 95% CI 0.010–0.063) than the composite PNI ($\Delta C\text{-stat} +0.022$, 95% CI -0.001 –0.051). Lymphocyte count alone provided negligible discrimination ($\Delta C\text{-stat} +0.002$, 95% CI -0.007 –0.014). A likelihood ratio test confirmed that adding lymphocyte count to an albumin-only model did not improve fit ($p=0.46$), indicating that lymphocyte count contributes no independent prognostic information beyond albumin in this population.

4. Discussion

4.1. Summary of Principal Findings

In this population-based dual-cohort analysis spanning 30 years of NHANES data, we demonstrate that CONUT and GNRI consistently predict all-cause mortality in GI cancer patients after Bonferroni correction, while PNI showed a nominally significant but Bonferroni-non-significant association ($p=0.020$) that requires replication. GNRI showed the strongest association (HR 0.67 per 1-SD, 95% CI 0.57–0.80), followed by CONUT (HR 0.71, 95% CI 0.61–0.83) and PNI

(HR 0.78, 95% CI 0.63–0.96). The PNI effect was strongly time-dependent, maximal in the first 2 years (HR 0.34), and operated primarily through non-cancer mortality rather than GI cancer-specific pathways. Critically, the prognostic value was driven by serum albumin alone, with lymphocyte count contributing no independent information. RMST analysis quantified a 1.97-year survival difference between high and low PNI groups at 10 years.

4.2. Time-Dependent and Cause-Specific Effects

CONUT and GNRI showed stable, time-independent associations with all-cause mortality across the full follow-up period. In landmark analyses, both remained significant at 1, 3, and 5 years, and the proportional hazards assumption was not violated for CONUT ($p=0.43$) or GNRI ($p=0.087$).

The temporal variation of PNI’s prognostic effect—maximal in the first 2 years (HR 0.34) followed by marked attenuation to null after excluding early deaths—stands in contrast to the stability of CONUT and GNRI. Pre-existing nutritional status most strongly influences short-term survival, likely through its impact on treatment tolerance, perioperative recovery, and therapy completion. This pattern aligns with prior observations in a Korean gastric cancer cohort [26] and suggests that the window for effective nutritional intervention is early in the cancer trajectory. The complete attenuation after 2 years (HR 0.98 in the >2-year sensitivity analysis) also suggests that a single baseline PNI measurement cannot capture the dynamic nature of nutritional status over extended follow-up.

The competing risks analysis provides mechanistic insight: PNI primarily protects against non-cancer death (HR 0.63, $p<0.001$) rather than GI cancer-specific death (HR 1.06, $p=0.70$), suggesting PNI captures physiological reserve and resilience to treatment-related complications rather than directly influencing tumor biology. This distinction is critical for clinical nutrition: it aligns with

370 ESPEN guidelines emphasizing that nutritional therapy aims to maintain capac-
371 ity for anti-cancer treatment and reduce non-cancer complications, rather than
372 to modify cancer biology per se [1, 2]. Patients with low PNI may benefit from
373 intensified supportive care and closer monitoring for non-cancer complications.
374 CONUT and GNRI showed protective effects against both cancer-specific and
375 non-cancer death, possibly reflecting the additional prognostic information con-
376 tributed by total cholesterol and BMI, which may capture aspects of metabolic
377 health relevant to cancer progression.

378 *4.3. Nutritional vs Inflammatory Pathways*

379 PNI decomposition revealed that serum albumin alone (HR 0.69, 95% CI
380 0.58–0.82) outperformed the composite PNI (HR 0.78) across multiple metrics:
381 lower AIC (1595.8 vs. 1607.8), larger C-statistic improvement, and significantly
382 better model fit (likelihood ratio test $p < 0.001$), while lymphocyte count showed
383 no independent prognostic value ($\Delta C\text{-stat} +0.002$, 95% CI -0.007 – 0.014). This
384 finding calls into question whether the lymphocyte component of PNI adds
385 meaningful prognostic information beyond albumin alone in this population.
386 The biological rationale for PNI’s association with mortality appears to be me-
387 diated primarily by albumin’s dual role as a nutritional marker and negative
388 acute-phase reactant reflecting systemic inflammation, rather than by the im-
389 munonutritional synergy PNI was originally designed to capture.

390 In the context of current consensus frameworks, PNI and CONUT may be
391 positioned as screening tools that identify patients warranting formal GLIM
392 (Global Leadership Initiative on Malnutrition) assessment. While PNI com-
393 ponents (albumin, lymphocyte count, total cholesterol) correlate with GLIM
394 etiologic criteria (disease burden/inflammation), they do not substitute for the
395 full GLIM phenotypic assessment including weight loss, low BMI, and reduced
396 muscle mass.

397 Inflammatory markers NLR and SII showed comparable prognostic perfor-
398 mance to PNI (+0.017 C-statistic improvement for both). The mutually ad-
399 justed model, in which neither PNI nor NLR remained significant, indicates
400 substantial overlap between nutritional and inflammatory signaling pathways.
401 The joint exposure analysis (Low PNI + High NLR, HR 1.67, 95% CI 1.06–2.63;
402 RERI –0.97) suggests that combined assessment identifies the highest-risk pa-
403 tients, though the negative RERI indicates overlapping rather than synergistic
404 pathways.

405 Notably, the absence of PNI $\times$ CRP interaction ($p=0.78$) and the non-
406 significant CALLY index finding (HR 0.85, $p=0.20$) indicate that PNI’s prog-
407 nostic value extends beyond what can be explained by systemic inflammation
408 measured through single biomarkers. The null dietary finding (HEI-2015 HR
409 0.97, $p=0.85$), though limited by reliance on a single 24-hour dietary recall
410 rather than usual intake estimates, reinforces the conceptual distinction be-
411 tween biochemical nutritional status—reflecting metabolic reserve and systemic
412 inflammation—and dietary intake patterns, which may be less relevant in the
413 post-diagnosis period when metabolic alterations dominate.

414 4.4. Comparison with Previous Literature

415 Our findings extend prior single-center studies validating PNI in GI can-
416 cer. A meta-analysis of 28 studies in gastric cancer reported that low PNI was
417 significantly associated with poorer overall survival (HR 1.38, 95% CI 1.23–
418 1.54) [9], consistent with our population-level results. More recently, analyses
419 using NHANES data have similarly demonstrated PNI’s prognostic value for
420 cancer-specific and all-cause mortality in US populations [27], corroborating
421 our findings in an independent sample. A novel contribution of our study is
422 the demonstration that the PNI effect is consistent across two NHANES gen-
423 erations (cohort interaction $p=0.48$), supporting generalizability across survey

424 eras. Our study uniquely provides dual-cohort cross-validation across 30 years,
425 absolute risk estimates (RMST) that enhance clinical interpretability, and time-
426 dependent analyses demonstrating the temporal window of nutritional risk. The
427 head-to-head comparison of PNI, CONUT, and GNRI—rarely performed in a
428 single population—identifies GNRI as the most stable and strongly associated
429 index, likely because BMI adjustment captures the paradoxical protective effect
430 of preserved body mass in cancer patients. Notably, the optimal PNI thresh-
431 old of 48.5 identified in this Western population is higher than the traditional
432 cutpoint of 40–45 reported in Asian surgical series [6, 12], possibly reflecting
433 differences in body composition, comorbidity burden, and healthcare context.
434 Cross-validation (mean 48.3) confirmed this threshold was not an artifact of
435 data-driven selection, though external validation in clinical cohorts is warranted.

436 4.5. *Clinical Implications*

437 These simple, inexpensive indices can be calculated from routine laboratory
438 data in virtually any clinical setting, aligning with ESPEN recommendations for
439 routine nutritional screening in cancer patients from the time of diagnosis [1].
440 GNRI and CONUT, which incorporate BMI or total cholesterol, showed more
441 stable, linear associations across the full follow-up period and may be prefer-
442 able for long-term risk stratification when these additional measurements are
443 available. A PNI threshold of 48.5 identifies a high-risk group with substan-
444 tially shorter survival (6.6 vs 12.8 years median survival; RMST difference 1.97
445 years at 10 years), though this finding should be interpreted cautiously given
446 PNI’s Bonferroni-non-significant primary result. The concentration of risk in
447 the first 2 years after assessment suggests that early nutritional evaluation and
448 intervention may have the greatest impact, consistent with the time-dependent
449 attenuation observed for single-timepoint measurements. In the context of the
450 GLIM framework, low values of nutritional indices may trigger formal malnutri-

tion assessment, which has clearer intervention pathways. The significant PNI $\times$ physical activity interaction ($p=0.023$) in our exploratory analysis suggests that the association between better nutritional status and survival may be enhanced by physical activity, a modifiable factor warranting further investigation in prospective studies.

4.6. *Strengths and Limitations*

Strengths include population-based sampling with national representativeness, dual-cohort cross-validation across 30 years, and comprehensive temporal analysis (time-dependent Cox, landmark, RCS, RMST, competing risks). We also performed head-to-head comparison of biochemical, dietary, and inflammatory indices, E-value sensitivity analysis, and bootstrap C-statistic comparisons with absolute risk measures (RMST difference) that enhance clinical interpretability. Extended covariate adjustment and cohort-stratified analyses demonstrated robustness to socioeconomic confounding and cross-generational consistency.

Several important limitations must be acknowledged. First, cancer stage, grade, and treatment data—the strongest prognostic factors in GI cancer—are not available in NHANES public-use data. The observed PNI-mortality association may partly reflect confounding by disease severity, particularly given that NHANES does not provide reliable data on time since cancer diagnosis (prevalent cohort design). Sensitivity analyses excluding early deaths partially address reverse causation but cannot substitute for stage-adjusted analysis. Second, cancer diagnoses were self-reported without medical record or cancer registry verification, which may introduce misclassification, particularly for upper GI sites.

Third, the use of two NHANES generations with different albumin assay methodologies (bromocresol green in NHANES III vs. bromocresol purple in Con-

tinuous NHANES) may introduce systematic measurement differences, though cohort-stratified analyses showed consistent effects and inclusion of cohort as a fixed effect did not alter results. Fourth, nutritional indices were measured at a single baseline time point, preventing assessment of longitudinal changes in nutritional status, which may be more clinically relevant than a single snapshot. Fifth, the public-use mortality file provides only a 10-group cause-of-death recode, limiting cause-specific analysis granularity.

Sixth, the sample composition (77% colorectal cancer) limits generalizability to upper GI malignancies, and site-specific subgroup analyses were underpowered (esophageal $n=15$, pancreatic $n=8$, gallbladder $n=6$). Seventh, the pooled analysis of NHANES III and Continuous NHANES spans different treatment eras; while formal interaction tests did not identify significant heterogeneity, secular changes in GI cancer management may introduce unmeasured confounding. Eighth, adjustment for only three demographic covariates in primary models leaves potential for residual confounding. However, extended models incorporating smoking status, a simplified Charlson comorbidity score, education, and income-to-poverty ratio—applied to the subset with available smoking data—did not attenuate the associations; PNI HR remained consistent (0.68, 95% CI 0.48–0.97, $p=0.035$), and CONUT and GNRI remained robust (all $p<0.001$). These findings suggest that the PNI-mortality association is not primarily driven by smoking or comorbidity confounding, though residual confounding by unmeasured factors (alcohol use, detailed frailty measures) cannot be excluded.

Ninth, the population attributable fraction estimate (13.4%) should be interpreted with caution as it assumes a causal relationship between low PNI and mortality, an assumption that cannot be verified in an observational design. Tenth, the HEI-2015 analysis relied on a single 24-hour dietary recall, which does not represent usual dietary intake and may bias associations toward the

505 null.

506 **5. Conclusions**

507 CONUT and GNRI are independently associated with all-cause mortality
508 in GI cancer patients after Bonferroni correction, with effects consistent across
509 two NHANES generations spanning 30 years. PNI showed a nominally signifi-
510 cant association that did not survive multiple testing correction and should be
511 interpreted as a secondary finding requiring independent replication. The pro-
512 tective associations are time-dependent (particularly for PNI, concentrated in
513 the first 2 years) and operate primarily through non-cancer mortality pathways,
514 suggesting these indices capture physiological reserve and resilience rather than
515 directly reflecting cancer-specific biology. An optimal PNI threshold of 48.5
516 identifies a high-risk subgroup with a 1.97-year RMST difference at 10 years,
517 though this requires validation. The prognostic value is driven predominantly
518 by serum albumin, with lymphocyte count contributing minimal independent
519 information, raising the question of whether albumin alone may suffice for risk
520 stratification. Inflammatory markers (NLR, SII) show comparable performance,
521 and joint assessment of PNI and NLR identifies a particularly high-risk group.
522 These simple, inexpensive indices may facilitate early risk stratification and trig-
523 ger formal nutritional assessment in GI cancer patients, though future studies
524 with cancer stage, treatment data, and longitudinal nutritional measurements
525 are needed to establish their independent prognostic value.

526 **Acknowledgments**

527 The authors thank the National Center for Health Statistics for conducting
528 the NHANES surveys and making the data publicly available.

529 **Data Availability Statement**

530 The data used in this study are publicly available from the National Health
531 and Nutrition Examination Survey (NHANES) at <https://www.cdc.gov/nchs/nhanes/>.
532 Analysis code is available from the corresponding author upon reasonable re-
533 quest.

534 **Funding Statement**

535 This research did not receive any specific grant from funding agencies in the
536 public, commercial, or not-for-profit sectors.

537 **Conflict of Interest**

538 The authors declare no conflicts of interest.

539 **Author Contributions**

540 **Zhuha Zhou:** Conceptualization, Methodology, Formal analysis, Investiga-
541 tion, Data curation, Writing—original draft, Writing—review & editing, Visu-
542 alization. **Yiqi Cai:** Supervision, Writing—review & editing. **Yongyu Bai:**
543 Investigation, Writing—review & editing. **Qigang Xu:** Data curation, Investi-
544 gation.

545 **Figure Legends**

- 546 1. **Figure 1:** Kaplan-Meier survival curves by PNI tertile (combined NHANES
547 cohorts).
- 548 2. **Figure 2:** Forest plot—adjusted HR per 1-SD for PNI, CONUT, and
549 GNRI.

- 550 3. **Figure 3:** Time-dependent HR across follow-up periods (0–2, 2–5, 5+
551 years).
- 552 4. **Figure 4:** Landmark analysis at 1, 3, and 5 years.
- 553 5. **Figure 5:** Competing risks forest plot—GI cancer death vs non-cancer
554 death.
- 555 6. **Figure 6:** PNI cutpoint analysis (optimal threshold identified at 48.5).
- 556 7. **Figure 7:** Restricted cubic spline dose-response curve for PNI.
- 557 8. **Figure 8:** Subgroup forest plot—PNI effect across strata.
- 558 9. **Figure 9:** PNI vs NLR vs SII comparison (C-statistic improvement).
- 559 10. **Figure 10:** PNI decomposition—albumin vs lymphocyte vs composite.

560 **Supplementary Materials**

561 **Tables S1–S7** and **Figures S1–S5** are available online as supplementary
562 data.

563 **References**

- 564 [1] M. Muscaritoli, J. Arends, P. Bachmann, et al., ESPEN practical guideline:
565 Clinical Nutrition in cancer, *Clinical Nutrition* 40 (5) (2021) 2898–2913.
566 doi:10.1016/j.clnu.2021.02.005.
- 567 [2] J. Arends, P. Bachmann, V. Baracos, et al., ESPEN guidelines on
568 nutrition in cancer patients, *Clinical Nutrition* 36 (1) (2017) 11–48.
569 doi:10.1016/j.clnu.2016.07.015.
- 570 [3] K. Fearon, F. Strasser, S. D. Anker, et al., Definition and classification of
571 cancer cachexia: an international consensus, *The Lancet Oncology* 12 (5)
572 (2011) 489–495. doi:10.1016/S1470-2045(10)70218-7.

- [4] X. Hébuterne, E. Lemarié, M. Michallet, et al., Prevalence of malnutrition and current use of nutrition support in patients with cancer, *Journal of Parenteral and Enteral Nutrition* 38 (2) (2014) 196–204. doi:10.1177/0148607113502674.
- [5] M. Pressoir, S. Desné, D. Berchery, et al., Prevalence, risk factors and clinical implications of malnutrition in French Comprehensive Cancer Centres, *British Journal of Cancer* 102 (6) (2010) 966–971. doi:10.1038/sj.bjc.6605578.
- [6] T. Onodera, N. Goseki, G. Kosaki, Prognostic nutritional index in gastrointestinal surgery of malnourished cancer patients, *Nihon Geka Gakkai Zasshi* 85 (9) (1984) 1001–1005.
- [7] J. Ignacio de Ulíbarri, A. González-Madroño, N. G. de Villar, et al., CONUT: a tool for controlling nutritional status. First validation in a hospital population, *Nutrición Hospitalaria* 20 (1) (2005) 38–45.
- [8] O. Bouillanne, G. Morineau, C. Dupont, et al., Geriatric Nutritional Risk Index: a new index for evaluating at-risk elderly medical patients, *The American Journal of Clinical Nutrition* 82 (4) (2005) 777–783. doi:10.1093/ajcn/82.4.777.
- [9] Y. Yang, P. Gao, Y. Song, et al., The prognostic nutritional index is a predictive indicator of prognosis and postoperative complications in gastric cancer: a meta-analysis, *European Journal of Surgical Oncology* 42 (8) (2016) 1176–1182. doi:10.1016/j.ejso.2016.05.029.
- [10] K. Sun, S. Chen, J. Xu, et al., The prognostic significance of the prognostic nutritional index in cancer: a systematic review and meta-analysis, *Journal of Cancer Research and Clinical Oncology* 140 (9) (2014) 1537–1549. doi:10.1007/s00432-014-1714-3.

- 599 [11] D. J. Pinato, B. V. North, R. Sharma, A novel, externally validated
600 inflammation-based prognostic algorithm in hepatocellular carcinoma: the
601 prognostic nutritional index (PNI), *British Journal of Cancer* 106 (8) (2012)
602 1439–1445. doi:10.1038/bjc.2012.92.
- 603 [12] K. Migita, T. Takayama, K. Saeki, et al., The prognostic nutritional in-
604 dex predicts long-term outcomes of gastric cancer patients independent
605 of tumor stage, *Annals of Surgical Oncology* 20 (8) (2013) 2647–2654.
606 doi:10.1245/s10434-013-2926-5.
- 607 [13] T. Nozoe, M. Kohno, T. Iguchi, et al., The prognostic nutritional index can
608 be a prognostic indicator in colorectal carcinoma, *Surgery Today* 42 (6)
609 (2012) 532–535. doi:10.1007/s00595-011-0061-0.
- 610 [14] M. Kanda, T. Fujii, Y. Kodera, et al., Nutritional predictors of postopera-
611 tive outcome in pancreatic cancer, *British Journal of Surgery* 98 (2) (2011)
612 268–274. doi:10.1002/bjs.7305.
- 613 [15] K. Han, I. Jung, J. H. Kim, et al., Restricted mean survival time for survival
614 analysis: a quick guide for clinicians, *Korean Journal of Radiology* 23 (6)
615 (2022) 561–568. doi:10.3348/kjr.2022.0061.
- 616 [16] P. Royston, M. K. Parmar, Restricted mean survival time: an alternative
617 to the hazard ratio for the design and analysis of randomized trials with
618 a time-to-event outcome, *BMC Medical Research Methodology* 13 (2013)
619 152. doi:10.1186/1471-2288-13-152.
- 620 [17] H. Putter, M. Fiocco, R. B. Geskus, Tutorial in biostatistics: competing
621 risks and multi-state models, *Statistics in Medicine* 26 (11) (2007) 2389–
622 2430. doi:10.1002/sim.2712.

- [18] P. K. Andersen, R. B. Geskus, T. de Witte, et al., Competing risks in epidemiology: possibilities and pitfalls, *International Journal of Epidemiology* 41 (3) (2012) 861–870. doi:10.1093/ije/dyr213.
- [19] J. P. Fine, R. J. Gray, A proportional hazards model for the subdistribution of a competing risk, *Journal of the American Statistical Association* 94 (446) (1999) 496–509. doi:10.1080/01621459.1999.10474144.
- [20] B. Rockhill, B. Newman, C. Weinberg, Use and misuse of population attributable fractions, *American Journal of Public Health* 88 (1) (1998) 15–19. doi:10.2105/AJPH.88.1.15.
- [21] T. J. VanderWeele, P. Ding, Sensitivity analysis in observational research: introducing the E-value, *Annals of Internal Medicine* 167 (4) (2017) 268–274. doi:10.7326/M16-2607.
- [22] D. Gupta, C. G. Lis, Pretreatment serum albumin as a predictor of cancer survival: a systematic review of the epidemiological literature, *Nutrition Journal* 9 (2010) 69. doi:10.1186/1475-2891-9-69.
- [23] Q. Tang, Q. Liu, Y. Zhou, et al., Predictive value of serum albumin levels on cancer survival: a prospective cohort study, *Frontiers in Oncology* 14 (2024) 1323192. doi:10.3389/fonc.2024.1323192.
- [24] A. J. Templeton, M. G. McNamara, B. Šeruga, et al., Prognostic role of neutrophil-to-lymphocyte ratio in solid tumors: a systematic review and meta-analysis, *Journal of the National Cancer Institute* 106 (6) (2014) dju124. doi:10.1093/jnci/dju124.
- [25] M. Yang, S. Q. Lin, X. Y. Liu, et al., Association between C-reactive protein-albumin-lymphocyte (CALLY) index and overall survival in pa-

- 647 tients with colorectal cancer, *Frontiers in Immunology* 14 (2023) 1131496.
648 doi:10.3389/fimmu.2023.1131496.
- 649 [26] J. Y. Lee, H. I. Kim, Y. N. Kim, et al., Clinical significance of the prognostic
650 nutritional index in predicting postoperative gastric cancer survival, *Jour-
651 nal of Gastric Cancer* 17 (2) (2017) 139–150. doi:10.5230/jgc.2017.17.e16.
- 652 [27] L. Zhao, X. Zhang, Y. Li, et al., Association of prognostic nutritional index
653 with mortalities in American cancer survivors: a retrospective study, *Food
654 Science & Nutrition* 12 (4) (2024) 2880–2893. doi:10.1002/fsn3.3877.